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Review Article

Cognitive bias research in forensic science: A systematic review

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ABSTRACT

The extent to which cognitive biases may influence decision-making in forensic science is an important question with implications for training and practice. We conducted a systematic review of the literature on cognitive biases in forensic science disciplines. The initial literature search including electronic searching of three databases (two social science, one science) and manual review of reference lists in identified articles. An initial screening of title and abstract by two independent reviewers followed by full text review resulted in the identification of 29 primary source (research) studies. A critical methodological deficiency, serious enough to make the study too problematic to provide useful evidence, was identified in two of the studies. Most (n = 22) conducted analyses limited to practitioners (n = 17), forensic science trainees (n = 2), or both forensic science practitioners and students (n = 3); other analyses were based on university student or general population participants. Latent fingerprint analysis was examined in 11 studies, with 1–3 other studies found in 13 other disciplines or domains. This set of studies provides a robust database, with evidence of the influence of confirmation bias on analysts conclusions, specifically among the studies with practitioners or trainees presented with case-specific information about the "suspect" or crime scenario (in 9 of 11 studies examining this question), procedures regarding use of exemplar(s) (in 4 of 4 studies), or knowledge of a previous decision (in 4 of 4 studies). The available research supports the idea of susceptibility of forensic science practitioners to various types of confirmation bias and of the potential value of procedures designed to reduce access to unnecessary information and control the order of providing relevant information, use of multiple comparison samples rather than a single suspect exemplar, and replication of results by analysts blinded to previous results.

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1. Introduction

Cognitive bias is a broad term that includes a variety of processes that may lead to inaccurate judgments or interpretations; cognitive biases can affect memory, reasoning, and decision-making [1]. Although the term may be interpreted with a negative

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connotation, it is important to understand that these processes are a natural consequence of the need to attune to patterns and develop heuristics to process a variety of complex stimuli [2]. Confirmation bias, described in 1998 [3] as "seeking or interpreting evidence in ways that are partial to existing beliefs, expectations, or a hypothesis in hand," is one type of cognitive bias with the potential to undermine the objective evaluation of forensic evidence. This could arise, for example, by an unconscious focusing of attention on similarities and away from dissimilarities because of information about other evidence in a case, attributes of the comparison procedures used, or from previous conclusions drawn by another analyst.

Interest in the extent to which cognitive biases may influence decision-making in forensic science has grown in the past decade; this was one of the research needs identified in the 2009 National Academy of Sciences report on forensic sciences [4]. The topic has been the focus of considerable discussion, as exemplified by the 2013 review by Kassin et al. [5] and the accompanying responses [2,6–14]. Restricting access to task-irrelevant information and controlling the order and time in which information is provided have been proposed as methods to reduce the potential for bias [15]. A recent survey of 403 expert forensic science examiners, however, found a general lack of acceptance of the need for procedures designed to minimize cognitive biases, and a failure to recognize susceptibility to biases [16].

We conducted a scoping review [17] of the literature on cognitive biases in criminal investigations and prosecutions to map the landscape of the existing research (e.g., topics, populations, variables), by the point in the criminal justice process being addressed. Forensic science was one of the areas with the most research relating to cognitive biases, and thus we focused on this set of research for a more in-depth analysis. In this second phase, we conducted a systematic review of the scope, design, and results of the forensic science-cognitive bias research using a set of criteria defined *a priori*. The purpose of this review was to evaluate the basis for conclusions regarding the potential influence of contextual bias and other forms of bias on decision making in different forensic science disciplines

2. Methods

We began our review using PsycINFO and Social Sciences Full Text, two electronic databases of social science articles, chapters, books, and dissertations. Specifically, we searched for work with the words (criminal or justice or police or investigation* or forensic* or jury or juries or judge* or conviction*) in any field (e.g., text, title) and ("cognitive bias" or "implicit bias" or "cognitive dissonance" or "tunnel vision" or "confirmation bias" or "interpretive bias" or "belief perseverance" or "asymmetrical skepticism") in any field, which produced 890 results. Based on title and abstract review by two independent reviewers, we identified 92 seemingly relevant abstracts, 20 of which pertained to forensic science. We also reviewed the reference lists within the identified forensic science publications to identify additional relevant references. As additional checks on the completeness of our search strategy, we repeated the search using the PubMed database, and our list of references was reviewed by three external researchers working on different questions relating to forensic science and confirmation bias. The last review of the reference list was in July, 2018. At the suggestion of a journal reviewer, we expanded our search to include studies of the interaction of humans with technology and databases, using a combination of search terms relating to bias, technology, and database systems. This search strategy identified a total of 41 primary source publications pertaining to 36 studies.

This review focuses on studies of the decisions or conclusions made in an analysis. Specifically, we searched for studies, in any language, addressing the following questions: (1) Does contextual or case information influence the results of an analysis? And (2) Does expectation bias (i.e., comparison to a single sample versus comparison to a set of samples; knowledge of a previous conclusion) influence the results of an analysis? These were issues that were frequently discussed in the commentaries and review articles found in the literature search. We limited our review to disciplines considered within the Organization of Scientific Area Committees for forensic science (OSAC) [see https://www.nist.gov/topics/forensic-science/scientific-area-committees]. This criterion was an easily reproducible, objective method to define the scope of our review, and was broad enough to include a wide variety of disciplines. This inclusion criterion resulted in the exclusion of 5 studies focusing on adversarial allegiance in forensic psychology [18–21] or medical expertise in the context of medical malpractice [22]. We included published studies and unpublished reports, but excluded one citation for which a full text report was not available [23]. Although all of the studies identified were in English, language had not been designated as an exclusion criterion. To avoid duplication of data, we did not include a meta-analysis based on two previous studies [24], but we did include the two underlying studies in our analysis. These exclusions resulted in a final set of 34 primary source publications pertaining to 29 studies for review (Table 1).

A systematic review would typically include an evaluation of the methods used in the identified studies to determine the "risk

Table 1
Summary of 27 studies of confirmation bias in forensic sciences.

Discipline or domain	N studies	Participants (Type, N)		N samples	Variable(s)
Fingerprints	8	Practitioner	5–70	1–>300	Case information, previous conclusion, comparison procedures, emotion, complexity
	3	University students	27–107	20–96	
Forensic Anthropology	3	practitioners, students	38–99	1	Case information
Bitemark	1	Dental students, other students	178	15	Complexity, emotion
Bloodstain	1	Practitioners	27; 30	12–16	Case information
Dog handling	1	Practitioners	18	8	Case information
DNA	1	Practitioners	17	1	Case information
Hair	1	Forensic science trainees	12	4	Comparison procedures
Handwriting	3	Forensic science trainees, practitioners, general population	12–28	6–30	Case information, previous conclusion, comparison procedures
Shoeprint	1	Practitioners	12	8	Case information, complexity
Speech (auditory evidence)	1	University students	145	17	Case information, complexity
Toolmarks (bullets)	1	Practitioners	6	6	Case information
Crime scene investigation	1	Practitioners, students	58, 36	1	Case information
Forensic pathology	1	Practitioners	192	31	Case information
Technology-human interactions	2	Practitioners, students	23–30	160–210	Previous (technology-based) conclusion

of bias," or the potential that methodological limitations would affect the interpretation of the results. Evaluation methods have been developed and used for randomized clinical trials and observational studies in medicine [25,26], policy initiatives in education, development, and other social sciences (see www.campbellcollaboration.org), and environmental health [27,28]. We drew from this previous work four domains relating to the internal validity of the study (allocation process or comparability of groups, blinding of participants, blinding of other outcome assessors, and analysis and reporting of results) and one domain relating to external validity or generalizability (participant eligibility and selection). The purpose of this evaluation was to (1) provide descriptive information pertaining to the extent each of these domains is addressed in this area of forensic science — cognitive bias research, and (2) as is recommended in systematic reviews, to identify attributes of the design (including implementation and analysis) that would be considered a critical deficiency, making the study too problematic to provide useful evidence. Exclusions for critical deficiencies can occur in the identification phase, through the delineation of specific exclusion criteria (e.g., excluding clinical trials that did not involve randomized allocation) [25] or in the evaluation phase, through the inclusion of a "critical risk of bias" category for the domains under review [26]. We began with a description of what would be considered an ideal or target study design, as discussed by Sterne et al. [26] and Hernán et al. [29]. We next considered design attributes within each domain that would be considered a critical deficiency, and then developed examples for intermediate levels (i.e., between "ideal" and "critical deficiency") of decisions or information that would strengthen the study (Appendix). In this way we were able to assess the studies' methods on a spectrum from ideal to deficient, focusing on suggestions that could help authors, and reviewers, strengthen the utility of research reports. Critical limitations are noted in the summary of the studies. To summarize the studies, we first stratified by type of research question addressed and discipline, and then abstracted the results for review. All studies, including those conducted with university student or general population participants, are included in the tabular summaries (Tables 2–5), but the synthesis focuses on studies of forensic science practitioners or trainees. We also provide a summary of the areas of improvement we noted in the study evaluations.

3. Results

3.1. Description of studies

The 29 forensic science-confirmation bias studies identified through this search included 11 studies of fingerprint (latent print) analysis, with 1–3 studies in each of 13 other disciplines or domains (Table 1). Most (n = 22) conducted analyses limited to practitioners (n = 17), forensic science trainees (n = 2), or both forensic science practitioners and students (n = 3); other analyses were based on university student or general population participants. Sample size was highly variable, ranging from 5 to 145 participants and from 1 to >300 samples to be analyzed per participant. The studies examined the influence of various types of case information or procedural details on decision-making; several studies also examined the interaction between complexity and case information. The two studies in the technology-domain examined the influence of software generated information (i.e., position in AFIS candidate list or classification using facial recognition software) on decision-making [30,31]. After the initial two studies were published in the 1980's [32,33], almost twenty years passed before the additional studies were conducted, with most published since 2010.

3.2. Synthesis of studies

Studies examining the influence of contextual (case) information on decision-making are summarized in Table 2. Examples of the type of information include information on whether there had been a confession, results from other analyses, and other details regarding the crime. Eleven disciplines are represented, most with only one study, but there were three studies in forensic anthropology [34–36]. Six experimental studies among forensic science practitioners demonstrated the influence of either the presence or absence of case information on analysts' conclusions. This included studies of analysis of DNA mixtures [37], fingerprints [38], bloodstain patterns [39–41], dog handling [42], crime scene investigation [43,44], and forensic pathology [45–47]. This effect was also seen in three studies using participants with variety of experience levels in forensic anthropology [34–36]. Two other studies, in shoeprint analysis [48] and bullet analysis [49] did not observe this type of influence. One study was considered to have critical deficiencies, due to the lack of comparability between groups [50]. The most relevant data in this study was the comparison among the 51 crimes against people among the 845 cases from 2009 to 2010, divided into 13 cases defined as high level of contextual information and high level of interaction with investigators and 38 cases with low levels of contextual information and low level of interaction with investigators. Residual confounding by type of crime within this broad category is likely, and there may be other differences between these groups that were not assessed or addressed in the analysis.

Four studies examined procedures relating to the type of exemplar or comparison sample(s) used in the analysis (Table 3). Two studies examined use of more than one comparison sample in microscopic hair analysis [33] and handwriting analysis [32]; the hair analysis study compared use of a "line-up" of 6 samples to use of a single exemplar, and the handwriting study compared a single suspect with three possible suspect scenarios. In both of these studies, the number of incorrect "identification" decisions was higher when only a single comparison sample was used. Two other studies examined assessment of minutia or suitability decisions in fingerprint analysis made with and without a target exemplar [51,52]. These two studies demonstrated an impact of a target exemplar on decision-making, with a higher number of minutia assessed with the presence of a target exemplar [51] and with the presence of an exemplar, particularly a non-matching exemplar, decreasing the correlation between the examiner's decision and the predetermined expert decision regarding suitability [52].

Six studies examined another procedural feature, specifically the influence of knowledge of a previous decision, on an analyst's conclusions (Table 4); one of these was critically limited by the failure of the blinding procedures, which was particularly difficult given the setting of this study outside of a workplace [53]. Three studies in fingerprints [52,54] and handwriting [55] were conducted among experienced practitioners; one of the technology-based studies was conducted with experienced latent print examiners [30] and the other technology-based study was conducted with university students [31]. Each of these studies provide evidence that being told or shown a decision, ostensibly based on an analyst's work or on a computer algorithm, resulted in conclusions that were more likely to be in agreement with the previous decision, even if the initial decision was incorrect.

Four studies examined emotional context or seriousness of crime, three in fingerprints [56–58] and one in bitemarks [59], but only two were conducted in experienced practitioners [57,58] (Table 5). Results for these two studies of suitability decisions among fingerprint examiners were mixed, with one study showing no difference in decision making [58] and the other showing a higher number of "false positive" decisions (a submission decision

Table 2
Studies of influence of case information on analyst decision-making, by discipline [64–66].

Discipline, Reference, n	Design Details	Results
DNA - mixtures		
Dror and Hampikian, 2011 [37] 17 practitioners, each examined 1 DNA mixture electropherogram	Between-person design. "Context free" analysis compared to previous conclusion of "cannot be excluded" by other examiners (n not reported) who had prosecution-relevant information. Outcome: Exclusion conclusion.	16 out of 17 context free conclusions differed from previous conclusion: 4 "inconclusive;" 12 "exclude;" 1 "cannot be excluded"
Fingerprints		
Dror and Charlton, 2006 [38] 6 practitioners, each given 8 of own previous cases decisions	Within-person design. 4 of the 8 samples given new contextual information ("suspect confessed;" "suspect was in custody"); varied by case complexity. Outcome: Identification conclusion.	Different decision made in 16.6% of cases with new information compared with 8.3% of cases without new information
Langenburg et al., 2014 [50] Crimes against people: 13 high context-high law enforcement and 38 low context-low interaction cases	Observational design. 51 crimes against people among 845 cases from 2009-2010. Compared results by level of information and law enforcement interaction. Outcome: Identification conclusion.	High context-High Interaction (n=13) / Low context-Low interaction (n=38) N (%) identifiable prints: 106 (8.2) / 20 (1.5) N (%) prints identified to suspect: 44 (1.2) / 8 (0.2) Critical limitation: Comparability of cases-residual confounding within "Crimes against people"
Smalarz et al., 2016 [64] 225 university students, each examined 1 mock crime incident report, suspect profile, and fingerprint	Between-person design. Varied type of crime (child molestation, identity theft) and suspect profile (based on pre-tested stereotypes: middle-aged White man; middle-aged Asian woman). Asked to compare crime scene fingerprint to suspect fingerprint. None were matches. Outcome: Match, do not match, inconclusive conclusion. [If "inconclusive" choice, forced decisions for match and do not match]	18.2% initially "inconclusive;"did not vary by crime-profile group. 34.2% "match" conclusions (including forced decisions). Crime type-suspect profile interaction (Wald test in logistic regression model = 3.63, p = 0.057), with difference seen for the child molestation crime: Child molestation White male 51.9% Asian woman 27.1% (p = 0.007) Identify theft [percentages from Figure 1] White male ~ 28% Asian woman ~ 28% p = 0.932)]\
Stevenage and Bennett, 2017 [65] 48 university students, with 20 minute ACE-V training and 8 practice trials, each examined 72 prints	Within- and between-person design. Varied within participants: information on DNA results (DNA match, DNA unclear, DNA does not match) and match status (36 matching and 36 non-matching prints). Varied between-participants: time pressure (2 seconds, unlimited). Outcome: Response bias (tendency to report either matches or non-matches) calculated as midpoint between standardized scores for "hits" (correct matches) and "false alarms" (incorrect match for a non-match)	Lower accuracy with time pressure and in non-match trials. 3-way interaction between trial type (match vs non-match)-time pressure-contextual information in analyses by[1]: Participant $F_{2,46}$ (p-value) / Item $F_{2,70}$ (p-value) Matching – No Time Pressure: 1.69 (0.20) / 1.92 (0.15) Matching – Time Pressure: 4.35 (0.019) / 4.78 (0.011) Non-matching – No Time Pressure: 4.78 (0.013) / 2.76 (0.070) Non-matching – Time Pressure: 8.53 (< 0.001) / 9.75 (< 0.001)
Forensic Anthropology		
Nakhaeizadeh et al., 2014a [34] 41 practitioners and students, each examined 1 set of skeletal remains	Between-person design. Varied information on DNA-indicated sex, origin, age at death. Conducted at a museum learning center. Outcome: Demographic characteristics conclusions.	Assessment of sex, ancestry, age at death strongly influenced by other information given, e.g.: n reporting Given DNA information / male / female / undetermined - indicating male: 10 / 2 / 2 - indicating female: 0 / 14 / 0 - no information: 4 / 9 / 0
Nakhaeizadeh et al., 2014b [35] 99 practitioners and students, each examined 14 images	Between-person design. Varied contextual information (mass grave; archeology site; control-no information). Varied complexity: 5 no trauma, 3 clear trauma, 6 ambiguous. Conducted through online website. Outcome: Level of trauma conclusion.	Mean rank trauma score given information indicating: - mass grave: 754.60 (p < 0.05 versus other groups) - archeology site: 651.42 - control 674.48

Nakhaeizadeh et al., 2016 [36] 38 Master's students, each examined 1 male skeletal cast	Between-person comparison. Varied contextual clothing (female or gender neutral clothing at the mock crime scene or controls-no clothing context information). Conducted as part of forensic anthropology course.	Clothes Context Group: Conclusion / Female (n=11) / Gender Neutral (n=12) / Controls (n=15) Male: 1 / 9 / 15 Possibly male: 2 / 2 / 0 Undetermined: 5 / 1 / 0 Possibly Female: 3 / 0 / 0 Female: 0 / 0 / 0

Bloodstain Pattern

Laber et al., 2014, Taylor et al., 2016a,b [39–41] 27-30 practitioners, separate studies for fabric and rigid surface, each examined 12-16 patterns	Between-person design. Varied contextual information (supporting correct answer, misleading, or neutral). Four patterns, three extent levels. Outcome: Type of pattern conclusion	% incorrect with: / Fabric / Rigid - supporting info: 14% / 8% - neutral info: 26% / 11% - misleading info: 30% / 20%

Dog Handling

Lit et al., 2011 [42] 18 handlers (for drug or explosive detector dogs), each completed 8 trials (two searches in four search areas)	Within-person design. Varied handler information (specific marker indicating a scent location), and decoy scents (Slim-Jim sausage, new tennis ball). Handler and observer recording the dog alerts were blinded to search conditions. Outcome: False alerts (by design, all alerts were false alerts)	15% of trials were "clear runs;" 85% had ≥ 1 false alert. Clear runs more likely in unmarked trials than in marked trials (n=15 and 6, respectively, p=0.05), indicating human influence. Clear runs did not differ in trials with and without decoy scent (n=11 and 10, respectively, p=0.83), indicating absence of dog influence.

Handwriting

Kukucka and Kassin, 2014 [66] 169 adults (general population), each examined 2 handwriting samples (robbery note and suspect sample)	Between-person design. Varied contextual information (confession-recantation or no confession) and similarity of the handwriting comparison samples (high-low). Conducted online. Outcome: 10-point scales for sample similarity rating, match conclusion, and guilty-not guilty conclusion	Total / Confession / No confession Similarity score: 4.17 (± 2.30) / 4.47 (± 2.21) / 3.86 (± 2.36) standardized difference (95% CI): 0.27 (-0.07, 0.61), p=0.085 Match conclusion: 18.93% / 26.75% / 10.84% odds ratio (95% CI): 3.00 (1.30, 6.96), p=0.008 Guilt conclusion: 15.38% / 24.42% / 6.02% odds ratio (95% CI): 5.04 (1.80, 14.1), p=0.001 Stronger confession effects for match and guilt conclusions in low similarity score compared with high similarity score condition

Auditory Evidence

Lange et al., 2011 [61] 145 university students, each listened to 1 recording (4 times) and recorded what they heard	Between-person design. Varied contextual information ("criminal suspects, job interview candidates, or no information) and degradation level of the recording (none, some-1000 Hz, high-670 Hz). Recorded statements consisted of 17 statements (6-14 word sentences) Outcome: correct transcription, suspect misinterpretations, job interview misinterpretations, (other) transcription errors (determined independently by two blinded raters, Kappa = 0.957)	For no degradation and highest degradation levels, transcription errors were not related to contextual information. For some degradation (1000 Hz), suspect misinterpretation was more likely compared with control interviews and job interviews. Likelihood Ratio (95% CI) Compared to controls (no information): 4.56 (2.34, 8.89) Compared to "job" interview: 4.15 (2.18, 7.90)

Shoeprint

Kerstholt et al., 2007 [48] 12 practitioners, each examined 8 (4 simple; 4 complex) cases (using manipulated photos)	Within-person design. Varied contextual information (e.g., details suggesting guilt). Examiners used standard Dutch guideline. All completed within 3 hour session, at workplace. Outcome: Numerical evidential value score of acquired feature(s).	Scores did not differ by inclusion of incriminating information: $F_{(1,6)} = 1.28$, $p > 0.3$ (magnitude, direction of results not reported). Scores were higher in simpler cases ($F_{(1,6)} = 8.44$, $p < 0.05$)

Toolmarks (bullets)

Kerstholt et al., 2010 [49] 6 practitioners, each examined 3 sets from same firearm and 3	Within-person design. Varied contextual information ("biasing" information indicated one gun). At least 6 months between assessments.	Variability between assessments seemed random rather than directional: With biased info, Conclusions / Same Barrel / Different Barrel

Discipline, Reference, n	Design Details	Results			
sets from different firearm [all same type]	Outcome: Same gun conclusion	- no change - more toward "same" - more toward "not same"	14 2 2	12 3 3	
Crime Scene Investigation					
van den Eeden et al., 2016, 2018 [43,44] 58 crime scene investigators, 36 students, each examined 360° panoramic views of ambiguous mock crime scene (hanging; murder staged as suicide). Given 30 minutes for computer-based investigation.	Between-person design. Varied contextual information (death considered supposed suicide - history of depression, witness saw nothing unusual; death considered supposed murder - history of domestic violence, witness saw man leaving the house; no information- no witnesses) Outcome: First impression of likely scenario (murder or suicide), number and type of secured traces, and post-processing impression of likely scenario		Contextual Information Group Suicide (n=19)	Murder (n=19)	No Information (n=20)
		First impression			
		- most likely suicide	89%	55%	79%
		- most likely murder	0%	5%	0%
		- indecisive	11%	40%	21%
		Number of secured traces:			
		Mean (± SD)	12.9 (±4.9)	17.8 (±6.6)	15.3 (±6.5)
		Post-processing impression			
		- most likely suicide	47%	30%	37%
		- most likely murder	26%	60%	42%
		- indecisive	26%	10%	21%
Forensic pathology					
Oliver et al., 2017a,b; Oliver and Fang, 2016 [45–47] 192 members of National Association of Medical Examiners (98 had completed a previous study with the same images; 90 had not completed the previous survey, 4 unknown status)	Between-person design. [Although more than half of the participants had also completed the first test, the analysis did not indicate that a paired comparison was used when limiting the analysis to this group.] Varied contextual information (case history information added to this study) Outcome: Level of consensus comparing current study to previous study			With addition of case information	Without case information
		Level of consensus: Mean (± SE) Median		91.3% (± 0.23) 97.9%	76.2% (± 0.03) 80.0%

[a]"not significant" designation of the authors replaced with actual p-value (http://www.socscistatistics.com/pvalues/fdistribution.aspx); p-values are 2-sided.

Table 3
Studies of Influence of Comparison Procedures on Analyst Decision-making, by Discipline.

Discipline, Reference, n	Design Details	Results
Fingerprints		
Dror et al., 2011 [51] 20 practitioners, each examined 10 latent marks	Within-person design. 5 with target exemplar and 5 with no comparison sample. Outcome: Minutia assessed.	Higher number of observed minutia with presence of target exemplar (p =0.021): With Exemplar / No Exemplar Mean (± SD) 12.7 (±4.7) / 10.1 (±2.0)
Fraser-Mackenzie et al., 2013 [52] Experiment 1: 24 practitioners, each presented with > 300 prints	Within-person design. > 125 prints with no exemplar and with > 180 prints + exemplar (matching or non-matching). Outcome: Suitability decision.	Correlation between predetermined decision regarding suitability and examiner's decision decreases with presence and type of exemplar: Spearman ρ = 0.795, 0.755, and 0.650, respectively, for no exemplar, matching exemplar, and non-matching exemplar (p < 0.001 for comparisons with no exemplar group)
Hair		
Miller, 1987 [33] 14 trainees (met requirements for expert testimony), each examined 4 samples	Within-person design. Varied procedure: 2 samples used standard procedure (comparison to samples from a single suspect) and 2 used "line-up" procedure (comparison to samples from 6 sources). None were same-source. Outcome: Incorrect conclusions (i.e., a "match" conclusion).	Incorrect conclusion made in 7 (30.4%) of standard procedure compared with 1 (3.8%) of "line-up" procedure samples (p < 0.01)
Handwriting		
Miller, 1984 [32] 12 trainees, each examined 1 case (with 3 checks)	Between-person design. Varied case procedure and case information (comparison to 1 "suspect," based on two witnesses, or to 3 "possible suspects"). None of the checks were written by any of the "suspects." Outcome: Incorrect conclusions (i.e., a "same source" conclusion).	More mistaken conclusions with single suspect comparison: Conclusion / Single suspect / 3 suspects Same source 4 0 Inconclusive 1 0 Did not write the check 1 6

Table 4
Studies of Influence of Knowledge of a Previous Conclusion on Analyst Decision-making, by Discipline or Domain.

Discipline or Domain, Reference, n	Design Details	Results
Fingerprints		
Dror et al., 2006 [54] 5 practitioners, each given own previous "match" case	Within-person design. Examiners told it was high profile exclusion case (Mayfield). Outcome: Identification conclusion.	3 changed:"match" to "no match" 1 changed: "match" to "can't decide" 1 did not change (still "match")
Langenburg et al., 2009 [53] 43 practitioners, each examined 6 unknown-exemplar prints	Between-person design. Varied information on previous conclusion (controls; low bias-anonymous expert; high bias-internationally-recognized expert). Outcome: Individualization-Inconclusive-Exclusion conclusion.	Inconclusives, by group: 7% in controls (n=15) 25% in low bias (n=12) 15% in high bias (n=16) Critical Limitation: Failure of participant blinding (participants "caught on")
Fraser-Mackenzie et al., 2013 [52] Experiment 2: 24 practitioners, each examined 90 fingerprints	Between-person design. Varied predetermined (expert) suitability category and display of previous conclusion. Outcome: Suitability conclusion.	Display of previous "Unsuitable" conclusion (regardless of the predetermined decision) resulted in more "unsuitable" decisions: Predetermined Conclusion Examiner's decision — Suitable — Unsuitable Suitable — 16.5% — 23.0% Questionable — 49.8% — 51.3% Unsuitable — 73.1% — 77.0% [p = 0.04]
Handwriting		
Merlino, 2015 [55] 49 practitioners and n=43 general population participants, each examined 11 signatures on two separate occasions (at least 1 day apart)	Within-person design. Varied second set information, either previous decision or manipulated previous decision. Outcome: Source conclusion [genuine, disguised (intentionally trying to make handwriting look different), or simulated (forged)].	From Table 3.5.3 in the report Among practitioners: Presenting a manipulated previous decision resulted in 60.0% (27 out of 45) changed decisions (non-matches) for disguised signatures and 21.4% (3 out of 14) non-matches for simulated signatures. Among general population: Presenting a manipulated previous decision resulted in 58.5% (79 out of 135) changed decisions (non-matches) for disguised signatures and 33.3% (14 out of 42) non-matches for simulated signatures.
Technology-Human Interaction		
Dror et al., 2012 [30] 23 practitioners (latent print examiners) Each examined 160 medium and low quality latent marks and prints (80 with 10-candidate lists and 80 with 20-candidate lists for a total of 2400 comparisons)	Within-person design. 10- or 20- candidate AFIS candidate lists generated from NIST database, with 5 different placements: - No match - Correct matching print at position 1 - Correct matching print at position 2 or 3 (for 10- and 20-candidate lists, respectively - Correct matching print at position 8 or 15 (for 10- and 20-candidate lists, respectively - Correct matching print at last position	False identifications most common in positions 1-3 (for 10-candidate lists) and positions 1-2 (20-candidate lists)
Fysh and Bindemann, 2018 [31] 3 experiments, each with 30 university students; 210 pairs of facial images (15 mismatched), presented in 3 sets of 70; labelled "same," "different," or "unresolved"	Within-person design within each experiment; between-person design for comparison across experiments. Experiment: 1) Told to ignore labels 2) Told that some labels were inaccurate, so they should check carefully 3) Immediate feedback on answers in first 2 sets; only 5 mismatched, in last set	1) Statistically significant "label" effect: Percent Correct by Label: Match — Mismatch Consistent — 81% — 69% Unresolved — 72% — 61% Inconsistent — 76% — 63% 2) Statistically significant "label" effect, "trial type" effect, and "label-trial type" interaction: Percent Correct by Label: Match — Mismatch Consistent — 85% — 70% Unresolved — 74% — 69% Inconsistent — 66% — 48% 3) Increased errors with feedback* Percent Correct by Label: Match — Mismatch Consistent — 96% — 60% Unresolved — 87% — 38% Inconsistent — 63% — 20% *Estimated from Figure 5

Discipline, Reference, n	Design Details	Results
Fingerprints		
Hall and Player, 2008 [58] 70 practitioners, each examined one latent print	Between-person design. Varied information about the crime: high emotional context (shooting) or low emotional context (forgery). Outcome: Identification – suitability conclusion.	Emotional Context: Conclusion / High / Low Identification 6 7 Not suitable for comparison 15 12 Not sufficient to identify 13 16 No identification 1 0 [Fisher p-value=0.76][2]
Earwaker et al., 2015 [57] 11 practitioners, each examined two sets of 20 prints	Within-person design. Varied type of crime: 1 set "serious crime and 1 set "volume crime" prints; 3 weeks between examinations. Outcome: "False positive = submission decision when expert decided insufficient quality; false negative = discard decision when expert decided sufficient quality.	Crime Context: Examiner Decision: / Serious / Volume Agreed with expert 139 152 False positive 50 26 False negative 31 42 [p < 0.001]

Table 5
Studies of Influence of Emotional Level or Seriousness of Case on Analyst Decision-making, by Discipline.

Discipline, Reference, n	Design Details	Results
Dror et al., 2005 [56] 27 university students, each examined 96 prints in 4 blocks of 24	Within-person design. Varied type of crime and information presentation: controls-no emotional influence, low emotional state (no physical harm, e.g., bicycle theft, burglary) with photographs; high emotional state (serious personal harm, e.g., murder, personal attack) with photographs, and high emotional state with photographs and subliminal message (e.g., "guilty", "same"). Also varied stimulation complexity (unambiguous match or non-match, ambiguous). Outcome: Same-Different conclusion	Stimulation-Emotional State Manipulation-interaction, with little difference in number of found matches in the unambiguous samples, but greater tendency to find matches (incorrect matches) with high emotional content conditions (p-value for t-tests versus controls): Unambiguous / Ambiguous Controls 50% 47% Low emotion 54% 49% High emotion 51% 58% (p = 0.05) High emotion + subliminal messaging 46% 66% (p = 0.012)
Bitemarks		
Osborne et al., 2014 [59] 178 dental students, 182 other students. Brief training and 6 practice trials. Each examined 96 bitemark samples (comparing photos of bitemark and dental cast overlays) in 3 blocks of 32	Within-person design. Varied emotional context (no information of high emotion-photo of violent crime) and subliminal message ("same" or "guilty"). Also varied ambiguity level (48 unambiguous, 48 ambiguous). Outcome: number of "match" conclusions (regardless of accuracy?)	No context effects seen in unambiguous samples. In ambiguous samples, matches increased with each trial when no additional information was provided, but declined slightly with the additional photo and subliminal message. Mean (± SE) Number of Matches In ambiguous samples: / No Context Group / Context Group No information 11.7 (± 0.18) 11.8 (±0.19) High emotion-pictures 12.1 (± 0.17) 11.7 (±0.21) High emotion pictures + subliminal messaging 12.3 (± 0.18) 11.7 (±0.22) [Data corresponds to Figure 3 of [59]; provided in personal correspondence, email July 7, 2018 from Dr. Niki Osborne]

[a]We report p-value for Fisher's 2x4 test (http://vassarstats.net/fisher2x4.html) because expected cell values are < 5; For Chi-square test, df for 4 x 2 table = 3.

made for samples in which a priori expert decision was insufficient quality) for serious (high emotional context) crimes [57].

3.3. Evaluation of study methods

As noted previously, we identified two studies with critical deficiencies relating to participant blinding and to comparability of groups. Among the other studies, design elements fell short of what we considered "ideal," either because important details were not reported, or because specific standards were not met. Although most studies used an experimental design with allocation of participants into study groups (or randomization of the ordering of samples for analysis), some of the studies did not specify that the allocation process was randomized, and none of the studies provided information about the randomization procedures, as is recommended for randomized clinical trials [see, for example, Chapter 8.9, Sequence Generation, in [60]]. In addition, randomization does not necessarily provide assurance of comparability, particularly for small sample sizes, and many studies did not report data on group-specific characteristics. For example, level of experience may be an important variable to consider in these analyses, and it is important to provide data on this variable for each of the study groups, rather than only for the full sample, to assure comparability across groups. The extent of blinding also varied among the studies, with some studies designed in a way that prevented participants from knowing that a particular sample was a study sample, and other studies only blinding participants to the specific hypothesis or to group allocation. Most studies reported descriptive analyses in enough detail to allow determination of the effect size (e.g., magnitude of differences between groups). More difficulty was seen with some of the specific statistical tests used, as methods may not have been described completely, or incorrect

procedures were used (e.g., use of Chi-square test with sparse cells). Most studies did not discuss eligibility criteria or details of the selection or recruitment process, including participation rates.

4. Discussion

The set of studies identified in this review provide evidence of the influence of contextual and confirmation bias on analysts' conclusions, specifically among the studies with practitioners or trainees examining case-specific information about the "suspect" or crime scenario (in 9 of 11 studies examining this question), procedures regarding use of exemplar(s) (in 4 of 4 studies), or knowledge of a previous decision (in 4 of 4 studies). The two studies among fingerprint analysts of the effect of emotional context or seriousness of a crime on suitability decision-making provide differing results; this may be a less important avenue for future research than the other questions examined.

The two studies that did not observe an influence of case-specific information on decision making were in shoe print and toolmark (bullet) analysis [48,49]. It is not possible to draw conclusions regarding differences in susceptibility to case-specific information among disciplines because the number of studies within a particular discipline is so small (in most cases, n = 1), and in many of these studies, the small sample size, in terms of participants, samples, or both, provides limited statistical power. If differences are established, it would be important to assess level of subjectivity in the analytical procedures as an explanatory factor.

The provision of case-specific information to forensic scientists raises several issues. As demonstrated in this set of studies, case-specific information can be influential, even when the information is wrong. Some types of information may be needed within the context of the analysis, however, so it is important to carefully consider whether, and when, information is provided to the analyst [15]. Another issue is that the use of ancillary information blurs the line between conclusions based on the attributes of the analysis of the specified evidence (e.g., the features of a latent print), which is the purview of the forensic scientist, and conclusions based on consideration of multiple lines of evidence, which is the purview of the trier of fact. The ancillary information may also give an illusion of a stronger basis for a conclusion than would be warranted by the analytical tools available to the forensic science practitioner.

The results of the studies examining aspects of comparison samples (target exemplars) provide insights regarding potential improvements to standard operating procedures. As has been suggested by Miller [33] and Wells et al. [14], the use of multiple comparison samples, analogous to using fillers in a photo array, may lead to a reduction in errors of identification.

The set of studies examining knowledge of a previous decision suggest additional procedural improvements. A verification step that is used only for some types of decisions creates a situation in which the second analyst has knowledge of the previous conclusion. If knowledge, or the perception of knowledge, of another analyst's conclusions influences decision-making, it would be important to develop procedures that allow for truly independent replication of results, for example through replication of all analyses, or of an appropriately weighted sample of analyses to prevent the second analyst from predicting or guessing the first analyst's decision. The influence of the results of an algorithm-generated matching process on human decision-making is another area requiring additional research.

Several studies examined, or at least discussed, a measure of complexity or sample quality as an effect modifier in their analysis of cognitive bias [38,45,48,52,56,57,59,61]. This is an important avenue to pursue, as the effects may be most pronounced in more complex or difficult analyses. Examination of effect modification requires attention to sample selection and allocation in the study design, and a larger sample size.

Our evaluation of study methods revealed areas in which improvements could be made to the design or the reporting of methods and results. Guidelines for reporting randomized controlled trials [62] and observational epidemiology studies [63], developed within the medical and health arena, may provide a useful foundation to think about ways to achieve, or come close to, an optimal design, and about the information that needs to be reported to allow readers to fully appreciate the strengths and limitations of the research. For example, even if the study will depend on practitioners who volunteer to participate, thinking through the population of interest, the recruitment process used for this population, and data that could be used to compare the sample to the population (or to compare participants and non-participants) may lead to decisions that result in stronger designs and more informative publications. If a study population is drawn from a single workplace, it would be useful to collect and report information on the total number of people eligible in the workplace (to be able to determine the participation rate), and relevant demographic and work-related variables that allow assessment of differences between eligible participants and non-participants. For studies with practitioners, it is important to incorporate study samples within the workplace flow or to conduct the study under workplace conditions. Allocation procedures, for example for grouping participants for between-person comparison designs, should be based on a randomization scheme, and the randomization process should be described. Blinding to the specific samples used in a study may be difficult, but possible, for within- and between-person designs; it can be useful to include in the design an assessment of the extent to which blinding was achieved.

Sample size is relevant to questions regarding precision, rather than bias; within-person comparisons can be more efficient compared with between-person designs because of they allow greater control of sources of variability. Although we did not designate an ideal sample size, or a size that would be a critical deficiency, we do note that many studies were quite small (<15 participants in total for within person comparisons, or per group). Larger studies would be better able to address interactions between study variables and provide a stronger basis for estimating effects even if methods to combine results across studies are used.

In summary, the available research supports the idea of susceptibility of forensic science practitioners to various types of cognitive bias, and the potential value of procedures designed to reduce access to unnecessary information and to promote use of multiple comparison samples rather than a single suspect exemplar and replication of results by analysts blinded to previous results. Future research should provide additional data in under-studied disciplines, assess level of subjectivity in the analytical procedures in relation to presence or magnitude of bias, and assess sample complexity or difficulty as an effect modifier. Attention to guidelines for designing and reporting studies may result in methodologically stronger, and more comprehensively described, studies.

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Appendix A. Study evaluation

1. Internal validity

Ideal design attributes	Critically deficient (high risk of bias)	Intermediate levels: what could be improved?
Allocation and comparability For randomized designs: • Randomization process described in detail; would not allow anticipation of allocation • Within-person comparison with randomized allocation of samples (sample selection or order); • Between person comparison with randomized allocation to groups; demographic and experience data indicate similarity of groups	For randomized designs: • Between person comparison with randomized allocation to groups; demographic and experience data indicate differences between groups that are not addressed in the analysis	For randomized designs: • Provide details of the randomization procedure; assure that participant or study personnel could not anticipate order or group assignment • Assess adequacy of randomization procedures with respect to comparability between groups and provide group-specific data for relevant characteristics • Assure selection process and experience of a "control" group differs only with respect to the specific intervention under investigation • Address relevant differences in the analysis and discussion
For non-randomized designs: • Relevant differences in case or person attributes adequately addressed in the analysis (consider processes leading to selection into the sample/groups)	For non-randomized designs: • Differences in case or person attributes not addressed in the analysis	For non-randomized designs: • Assess comparability of groups (potential confounding) in the analysis and discussion
Blinding – participants • Blinding to purpose-hypothesis-study question • Conducted as part of regular workplace or course activities (do not know samples are study samples) • Adequacy of blinding procedures assessed	• Evidence of failure of participant blinding	• Incorporate into workplace or course activities • Include assessment of adequacy of blinding procedures
Blinding – Others Interacting with Participants (including anyone making subjective interpretations of participants' results or decisions) • Blinded to participant group	• Evidence of failure of blinding, affecting participant behavior or assessors' decision-making	• To the extent possible, build in protections to prevent interactions with participants by people knowledgeable of hypothesis and group assignments • For any subjective assessment of participants' decisions, describe training, use two raters and assess level of agreement
Analysis and reporting of results • Data presentation or analysis provide effect size, direction and variability measure • Statistical procedures appropriate for the data • Relevant variables, influence of outliers and robustness of results addressed in the analysis • Outcome measure appropriately sensitive to the type of effect being assessed	• Data presentation or results did not provide effect size, direction and variability measure for the results • Inappropriate statistical tests, without adequate data presentation to allow correct testing to be conducted • Insensitive outcome measure (would not be able to ascertain an effect if an effect was present)	• Provide everything needed to interpret results (magnitude, direction, and variability in effect measures) • Assess adequacy of assumptions-requirements for statistical tests • Where applicable, considered relevant variables, influence of outliers and robustness of results • Make sure text matches tables-figures

2. External validity - participant eligibility and selection – if target population is experienced practitioners

In Order of Decreasing Generalizability to Target Population	• Limited to practitioners with specified minimum level of experience (e.g., 2 years) • Limited to practitioners, no minimum level of experience • Limited to practitioners, no information about level of experience • Students or trainees in the relevant forensic science discipline • General population or university students
Other important design attributes	• Eligibility criteria, sampling frame and recruitment-selection process specified • To the extent possible, participation rate at each step of recruitment process assessed and high participation rate (e.g., >95%) achieved for all steps with a specified sampling frame • Data pertaining to participant characteristics allow analysis of representativeness of study sample, and for selected characteristics, allow comparison between participants and non-participants

References

[1] A. Tversky, D. Kahneman, Judgment under uncertainty: heuristics and biases, Science 185 (1974) 1124–1131.
[2] R.N. Haber, L. Haber, The culture of science: bias and forensic evidence, J. Appl. Res. Mem. Cogn. 2 (2013) 65–67, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.005.
[3] R.S. Nickerson, Confirmation Bias: a ubiquitous phenomenon in many guises, Rev. Gen. Psychol. 2 (1998) 175–220.
[4] National Research Council, Strengthening Forensic Science in the United States: a Path Forward, National Academy of Sciences, 2009.
[5] S.M. Kassin, I.E. Dror, J. Kukucka, The forensic confirmation bias: problems, perspectives, and proposed solutions, J. Appl. Res. Mem. Cogn. 2 (2013) 42–52, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.001.

[6] L. Butt, The forensic confirmation bias: problems, perspectives, and proposed solutions—commentary by a forensic examiner, J. Appl. Res. Mem. Cogn. 2 (2013) 59–60, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.012.
[7] D. Charlton, Standards to avoid bias in fingerprint examination? Are such standards doomed to be based on fiscal expediency?, J. Appl. Res. Mem. Cogn. 2 (2013) 71–72, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.009.
[8] S.D. Charman, The forensic confirmation bias: a problem of evidence integration, not just evidence evaluation, J. Appl. Res. Mem. Cogn. 2 (2013) 56–58, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.010.
[9] S.A. Cole, Implementing counter-measures against confirmation bias in forensic science, J. Appl. Res. Mem. Cogn. 2 (2013) 61–62, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.011.
[10] I.E. Dror, S.M. Kassin, J. Kukucka, New application of psychology to law: improving forensic evidence and expert witness contributions, J. Appl. Res. Mem. Cogn. 2 (2013) 78–81, doi:http://dx.doi.org/10.1016/j.jarmac.2013.02.003.
[11] E. Elaad, Psychological contamination in forensic decisions, J. Appl. Res. Mem. Cogn. 2 (2013) 76–77, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.006.
[12] R. Heyer, C. Semmler, Forensic confirmation bias: the case of facial image comparison, J. Appl. Res. Mem. Cogn. 2 (2013) 68–70, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.008.
[13] M. Triplett, Errors in forensics: cause(s) and solutions, J. Appl. Res. Mem. Cogn. 2 (2013) 63–64, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.007.
[14] G.L. Wells, M.M. Wilford, L. Smalarz, Forensic science testing: the forensic filler-control method for controlling contextual bias, estimating error rates, and calibrating analysts' reports, J. Appl. Res. Mem. Cogn. 2 (2013) 53–55, doi:http://dx.doi.org/10.1016/j.jarmac.2013.01.004.
[15] I.E. Dror, W.C. Thompson, C.A. Meissner, I. Kornfield, D. Krane, M. Saks, M. Risinger, Letter to the editor- context management toolbox: a linear sequential unmasking (LSU) approach for minimizing cognitive Bias in forensic decision making, J. Forensic Sci. 60 (2015) 1111–1112, doi:http://dx.doi.org/10.1111/1556-4029.12805.
[16] J. Kukucka, S.M. Kassin, P.A. Zapf, I.E. Dror, Cognitive bias and blindness: a global survey of forensic science examiners, J. Appl. Res. Mem. Cogn. 6 (2017) 452–459, doi:http://dx.doi.org/10.1016/j.jarmac.2017.09.001.
[17] H. Arksey, L. O'Malley, Scoping studies: towards a methodological framework, Int. J. Soc. Res. Methodol. 8 (2005) 19–32.
[18] B.D. McAuliff, J.L. Arter, Adversarial allegiance: the devil is in the evidence details, not just on the witness stand, Law Hum. Behav. 40 (2016) 524–535, doi:http://dx.doi.org/10.1037/lhb0000198.
[19] D. Mossman, When forensic examiners disagree: bias, or just inaccuracy? Psychol. Public Policy Law 19 (2013) 40–55, doi:http://dx.doi.org/10.1037/a0029242.
[20] D.C. Murrie, M.T. Boccaccini, J.T. Johnson, C. Janke, Does interrater (dis) agreement on Psychopathy Checklist scores in sexually violent predator trials suggest partisan allegiance in forensic evaluations? Law Hum. Behav. 32 (2008) 352–362, doi:http://dx.doi.org/10.1007/s10979-007-9097-5.
[21] D.C. Murrie, M.T. Boccaccini, L.A. Guarnera, K.A. Rufino, Are forensic experts biased by the side that retained them? Psychol. Sci. 24 (2013) 1889–1897, doi:http://dx.doi.org/10.1177/0956797613481812.
[22] D.V. Yokum, Debiasing the Courtroom: using Behavioral Insights to Avoid and Mitigate Cognitive Biases, ProQuest Information & Learning, 2016.
[23] C. Semmler, The Impact of Individual Differences and Eye Movements on Facial Comparison Performanc, (2012) .
[24] I. Dror, R. Rosenthal, Meta-analytically quantifying the reliability and biasability of forensic experts, J. Forensic Sci. 53 (2008) 900–903, doi:http://dx.doi.org/10.1111/j.1556-4029.2008.00762.x.
[25] J.P.T. Higgins, D.G. Altman, P.C. Gotzsche, P. Juni, D. Moher, A.D. Oxman, J. Savovic, K.F. Schulz, L. Weeks, J.A.C. Sterne, Cochrane Bias Methods Group, Cochrane Statistical Methods Group, The Cochrane Collaboration's tool for assessing risk of bias in randomised trials, BMJ 343 (2011), doi:http://dx.doi.org/10.1136/bmj.d5928 d5928–d5928.
[26] J.A. Sterne, M.A. Hernán, B.C. Reeves, J. Savović, N.D. Berkman, M. Viswanathan, D. Henry, D.G. Altman, M.T. Ansari, I. Boutron, J.R. Carpenter, A.-W. Chan, R. Churchill, J.J. Deeks, A. Hróbjartsson, J. Kirkham, P. Jüni, Y.K. Loke, T.D. Pigott, C. R. Ramsay, D. Regidor, H.R. Rothstein, L. Sandhu, P.L. Santaguida, H.J. Schünemann, B. Shea, I. Shrier, P. Tugwell, L. Turner, J.C. Valentine, H. Waddington, E. Waters, G.A. Wells, P.F. Whiting, J.P. Higgins, ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions, BMJ (2016) i4919, doi:http://dx.doi.org/10.1136/bmj.i4919.
[27] G.S. Cooper, R.M. Lunn, M. Ågerstrand, B.S. Glenn, A.D. Kraft, A.M. Luke, J.M. Ratcliffe, Study Sensitivity: evaluating the ability to detect effects in systematic reviews of chemical exposures, Environ. Int. 92–93 (2016) 605–610, doi:http://dx.doi.org/10.1016/j.envint.2016.03.017.
[28] A.A. Rooney, G.S. Cooper, G.D. Jahnke, J. Lam, R.L. Morgan, A.L. Boyles, J.M. Ratcliffe, A.D. Kraft, H.J. Schünemann, P. Schwingl, T.D. Walker, K.A. Thayer, R. M. Lunn, How credible are the study results? Evaluating and applying internal validity tools to literature-based assessments of environmental health hazards, Environ. Int. 92–93 (2016) 617–629, doi:http://dx.doi.org/10.1016/j.envint.2016.01.005.
[29] M.A. Hernán, B.C. Sauer, S. Hernández-Díaz, R. Platt, I. Shrier, Specifying a target trial prevents immortal time bias and other self-inflicted injuries in observational analyses, J. Clin. Epidemiol. 79 (2016) 70–75, doi:http://dx.doi.org/10.1016/j.jclinepi.2016.04.014.
[30] I.E. Dror, K. Wertheim, P. Fraser-Mackenzie, J. Walajtys, The impact of human-technology cooperation and distributed cognition in forensic science: biasing effects of AFIS contextual information on human experts, J. Forensic Sci. 57 (2012) 343–352, doi:http://dx.doi.org/10.1111/j.1556-4029.2011.02013.x.
[31] M.C. Fysh, M. Bindemann, Human-computer interaction in face matching, Cogn. Sci. 42 (2018) 1714–1732, doi:http://dx.doi.org/10.1111/cogs.12633.
[32] L.S. Miller, Bias among forensic document examiners: a need for procedural change, J. Police Sci. Adm. 12 (1984) 407–411.
[33] L.S. Miller, Procedural bias in forensic science examinations of human hair, Law Hum. Behav. 11 (1987) 157–163, doi:http://dx.doi.org/10.1007/BF01040448.
[34] S. Nakhaeizadeh, I.E. Dror, R.M. Morgan, Cognitive bias in forensic anthropology: visual assessment of skeletal remains is susceptible to confirmation bias, Sci. Justice J. Forensic Sci. Soc. 54 (2014) 208–214, doi:http://dx.doi.org/10.1016/j.scijus.2013.11.003.
[35] S. Nakhaeizadeh, I. Hanson, N. Dozzi, The power of contextual effects in forensic anthropology: a study of biasability in the visual interpretations of trauma analysis on skeletal remains, J. Forensic Sci. 59 (2014) 1177–1183, doi:http://dx.doi.org/10.1111/1556-4029.12473.
[36] S. Nakhaeizadeh, R.M. Morgan, C. Rando, I.E. Dror, Cascading Bias of initial exposure to information at the crime scene to the subsequent evaluation of skeletal remains, J. Forensic Sci. (2017), doi:http://dx.doi.org/10.1111/1556-4029.13569.
[37] I.E. Dror, G. Hampikian, Subjectivity and bias in forensic DNA mixture interpretation, Sci. Justice J. Forensic Sci. Soc. 51 (2011) 204–208, doi:http://dx.doi.org/10.1016/j.scijus.2011.08.004.
[38] I.E. Dror, D. Charlton, Why experts make errors, J. Forensic Identif. 56 (2006) 600–616.
[39] T. Laber, P. Kish, M. Taylor, G. Owens, N. Osborne, J. Curran, Reliability Assessment Of Current Methods In Bloodstain Pattern Analysis, NIJ, 2014.
[40] M.C. Taylor, T.L. Laber, P.E. Kish, G. Owens, N.K.P. Osborne, The reliability of pattern classification in bloodstain pattern analysis - PART 2: bloodstain patterns on fabric surfaces, J. Forensic Sci. 61 (2016) 1461–1466, doi:http://dx.doi.org/10.1111/1556-4029.13191.
[41] M.C. Taylor, T.L. Laber, P.E. Kish, G. Owens, N.K.P. Osborne, The reliability of pattern classification in bloodstain pattern analysis, part 1: bloodstain patterns on rigid non-absorbent surfaces, J. Forensic Sci. 61 (2016) 922–927, doi:http://dx.doi.org/10.1111/1556-4029.13091.
[42] L. Lit, J.B. Schweitzer, A.M. Oberbauer, Handler beliefs affect scent detection dog outcomes, Anim. Cogn. 14 (2011) 387–394, doi:http://dx.doi.org/10.1007/s10071-010-0373-2.
[43] C.A.J. van den Eeden, C.J. de Poot, P.J. van Koppen, Forensic expectations: investigating a crime scene with prior information, Sci. Justice 56 (2016) 475–481, doi:http://dx.doi.org/10.1016/j.scijus.2016.08.003.
[44] C.A.J. van den Eeden, C.J. de Poot, P.J. van Koppen, The forensic confirmation bias: a comparison between experts and novices, J. Forensic Sci. (2018).
[45] W.R. Oliver, Effect of history and context on forensic pathologist interpretation of photographs of patterned injury of the skin, J. Forensic Sci. 62 (2017) 1500–1505, doi:http://dx.doi.org/10.1111/1556-4029.13449.
[46] W.R. Oliver, X. Fang, Forensic pathologist consensus in the interpretation of photographs of patterned injuries of the skin, J. Forensic Sci. 61 (2016) 972–978, doi:http://dx.doi.org/10.1111/1556-4029.13092.
[47] W.R. Oliver, Reasons for lack of consensus in forensic pathologist interpretation of photographs of patterns of injury of the skin, J. Forensic Sci. 62 (2017) 674–680, doi:http://dx.doi.org/10.1111/1556-4029.13322.
[48] J.H. Kerstholt, R. Paashuis, M. Sjerps, Shoe print examinations: effects of expectation, complexity and experience, Forensic Sci. Int. 165 (2007) 30–34, doi:http://dx.doi.org/10.1016/j.forsciint.2006.02.039.
[49] J. Kerstholt, A. Eikelboom, T. Dijkman, R. Stoel, R. Hermsen, B. van Leuven, Does suggestive information cause a confirmation bias in bullet comparisons? Forensic Sci. Int. 198 (2010) 138–142, doi:http://dx.doi.org/10.1016/j.forsciint.2010.02.007.
[50] G. Langenburg, F. Bochet, S. Ford, A report of statistics from latent print casework, Forensic Sci. Policy Manag. Int. J. 5 (2014) 15–37, doi:http://dx.doi.org/10.1080/19409044.2014.929759.
[51] I.E. Dror, C. Champod, G. Langenburg, D. Charlton, H. Hunt, R. Rosenthal, Cognitive issues in fingerprint analysis: inter- and intra-expert consistency and the effect of a "target" comparison, Forensic Sci. Int. 208 (2011) 10–17, doi:http://dx.doi.org/10.1016/j.forsciint.2010.10.013.
[52] P.A.F. Fraser-Mackenzie, I.E. Dror, K. Wertheim, Cognitive and contextual influences in determination of latent fingerprint suitability for identification judgments, Sci. Justice J. Forensic Sci. Soc. 53 (2013) 144–153, doi:http://dx.doi.org/10.1016/j.scijus.2012.12.002.
[53] G. Langenburg, C. Champod, P. Wertheim, Testing for potential contextual bias effects during the verification stage of the ACE-V methodology when conducting fingerprint comparisons, J. Forensic Sci. 54 (2009) 571–582, doi:http://dx.doi.org/10.1111/j.1556-4029.2009.01025.x.
[54] I.E. Dror, D. Charlton, A.E. Péron, Contextual information renders experts vulnerable to making erroneous identifications, Forensic Sci. Int. 156 (2006) 74–78, doi:http://dx.doi.org/10.1016/j.forsciint.2005.10.017.
[55] M. Merlino, Reliability Validity, Accuracy, and Bias in Forensic Signature Identification, NIJ, 2015.
[56] I.E. Dror, A.E. Péron, S.-L. Hind, D. Charlton, When Emotions Get the Better of Us: The Effect of Contextual Top-down Processing on Matching Fingerprints, Appl. Cogn. Psychol. 19 (2005) 799–809, doi:http://dx.doi.org/10.1002/acp.1130.
[57] H. Earwaker, R.M. Morgan, A.J.L. Harris, L.J. Hall, Fingermark submission decision-making within a UK fingerprint laboratory: Do experts get the marks

that they need? Sci. Justice J. Forensic Sci. Soc. 55 (2015) 239–247, doi:http://dx.doi.org/10.1016/j.scijus.2015.01.007.
[58] L.J. Hall, E. Player, Will the introduction of an emotional context affect fingerprint analysis and decision-making? Forensic Sci. Int. 181 (2008) 36–39, doi:http://dx.doi.org/10.1016/j.forsciint.2008.08.008.
[59] N.K.P. Osborne, S. Woods, J. Kieser, R. Zajac, Does contextual information bias bitemark comparisons? Sci. Justice J. Forensic Sci. Soc. 54 (2014) 267–273, doi:http://dx.doi.org/10.1016/j.scijus.2013.12.005.
[60] J.P. Higgins, S. Green, Cochrane Handbook for Systematic Reviews of Interventions, (2011) . http://handbook-5-1.cochrane.org/.
[61] N.D. Lange, R.P. Thomas, J. Dana, R.M. Dawes, Contextual biases in the interpretation of auditory evidence, Law Hum. Behav. 35 (2011) 178–187, doi:http://dx.doi.org/10.1007/s10979-010-9226-4.
[62] K.F. Schulz, D.G. Altman, D. Moher, CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials, BMJ 340 (2010), doi:http://dx.doi.org/10.1136/bmj.c332.
[63] J.P. Vandenbroucke, E. von Elm, D.G. Altman, P.C. Gøtzsche, C.D. Mulrow, S.J. Pocock, C. Poole, J.J. Schlesselman, M. Egger, STROBE Initiative, Strengthening the reporting of observational studies in epidemiology (STROBE): explanation and elaboration, Int. J. Surg. Lond. Engl. 12 (2014) 1500–1524, doi:http://dx.doi.org/10.1016/j.ijsu.2014.07.014.
[64] L. Smalarz, S. Madon, Y. Yang, M. Guyll, S. Buck, The perfect match: do criminal stereotypes bias forensic evidence analysis? Law Hum. Behav. 40 (2016) 420–429, doi:http://dx.doi.org/10.1037/lhb0000190.
[65] S.V. Stevenage, A. Bennett, A biased opinion: demonstration of cognitive bias on a fingerprint matching task through knowledge of DNA test results, Forensic Sci. Int. 276 (2017) 93–106, doi:http://dx.doi.org/10.1016/j.forsciint.2017.04.009.
[66] J. Kukucka, S.M. Kassin, Do confessions taint perceptions of handwriting evidence? An empirical test of the forensic confirmation bias, Law Hum. Behav. 38 (2014) 256–270, doi:http://dx.doi.org/10.1037/lhb0000066.